Just a Few Days Left!

When: Saturday, April 18th from 10 am - 12 am (MIDNIGHT)

Where: Sitterson and Fred Brooks 1st floor (Floor 0)

- Please confirm your attendance by responding to the official Hack110 email.
- Be on the lookout for an email to join the Hack110 GroupMe
- **Don't want to make a project or stay the full time?** We have plenty of workshops throughout the day to benefit you in multiple ways!
- Still time to sign up. Don't Wait!

Website!

Questions? Email **hack110@office.unc.edu**

Perks

Lunch catered by **Panera**

Dinner catered by **Kolapasi**

Free Hack 110 Shirt

Free Snacks and **Drinks** Throughout

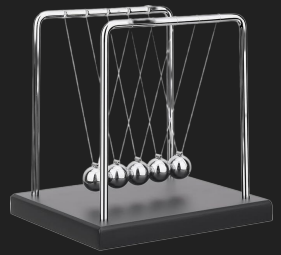
Earn **CLE credit**

Win Vouchers for UNC's best cafe: **Tea Hill**

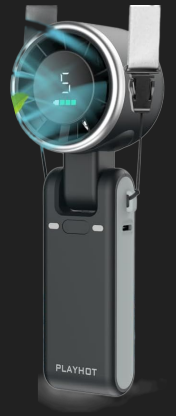
Compete to **win** many amazing **prizes!**

Select a Prize from the Pool

Laptop Stand

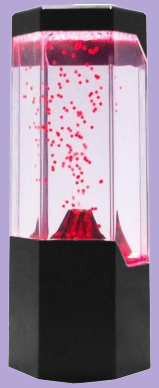


Newton’s Cradle



Portable Fan

Survival Kit



Volcano Lamp

Tile Tracker

Wireless Keyboard

Galaxy Projector

Power Banks

Phone Mount

COMP 110

Analyzing Data from a CSV in a Jupyter Notebook

Announcements

Re: Assignments:

- **EX08: Linked List Utility Functions** due Wed. April 15th

Quiz 03 this Thursday! To prep:

- Review your previous quiz(zes)
- Practice problems on the website
- Review session on Wednesday evening (details TBA soon!)

We've covered a LOT in this course thus far…

- Objects
- Data types
- Expressions
- Functions
- Memory diagrams
- Boolean expressions
- Conditionals
- Scope
- User input
- While loops
- For loops
- Importing modules
- Lists
- Unit tests
- Dictionaries
- Object-oriented prog.
- Classes and methods
- Magic methods + operator overloading
- Recursive structures
- Recursive functions/ methods
- (and more…)

Now, it's time to take these ideas and put them to use by analyzing real-world data!

EX09: Data Analysis for Continuous Improvement

For your last exercise, you will be analyzing the anonymized data from the class survey (LS16) to make evidence-based recommendations regarding how we can improve the course in the future!

This assignment will require you to:

we'll introduce these today!

- Read a CSV of data in Python
- Use "helper" functions to wrangle the data
- Analyze the data
- Generate figures summarizing the results of your analysis
  - (with the help of a library called **`seaborn`**!)
- Summarize your work and findings on your own website

New skills!

Let’s analyze Raleigh weather data and Walter Wally’s predictions to see how accurate his winter weather forecasts are!

(Walter Wally is Raleigh’s version of Punxsutawney Phil!)

Today’s agenda:

1. Introduce Jupyter Notebooks
    a. Markdown cells (formatted text)
    b. Python cells (code)
2. Review CSV structure
3. Read CSV data
4. Wrangle data

Next week:

1. Analyze data
2. Read documentation/use Python packages and libraries

What is a Jupyter Notebook?

An .ipynb file mixes live Python code, rich text, and output, all in one document. Instead of running an entire script at once, you execute small chunks (“cells”) and see output immediately below each cell.

Generate + Code + Markdown ··· .conda (Python 3.11.8)

Generate + Code + Markdown

```markdown
# Introduction to Jupyter Notebooks!

This is a Markdown cell (used to format plain text). See [this markdown guide](https://www.markdownguide.org/cheat-sheet/) for examples of markdown syntax that adjusts the format of plain text!
```

markdown

```python
# This is a Python cell.
name: str = "Alyssa"

print(f"{name} is my Prof!")
```

[1] ✓ 0.0s Python

```
Alyssa is my Prof!
```

In your VSCode workspace:

1. Create a directory (folder) called **`data_analysis`**
2. Inside **`data_analysis`**, create a new file called **`analysis.ipynb`**

What is a Jupyter Notebook?

Let's try it!

An .ipynb file mixes live Python code, rich text, and output, all in one document. Instead of running an entire script at once, you execute small chunks ("cells") and see output immediately below each cell.

Generate + Code + Markdown ··· .conda (Python 3.11.8)

Generate + Code + Markdown

```markdown
# Introduction to Jupyter Notebooks!

This is a Markdown cell (used to format plain text). See [this markdown guide](https://www.markdownguide.org/cheat-sheet/) for examples of markdown syntax that adjusts the format of plain text!
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```python
# This is a Python cell.
name: str = "Alyssa"

print(f"{name} is my Prof!")
```

[1] ✓ 0.0s Python

```
Alyssa is my Prof!
```

Using Notebooks in VS Code

1. **Open or create an .ipynb file**
   New file... → name it ***<anything>.ipynb***
2. **Select your Python kernel**
   Click the kernel picker (top right) → select a Python version
3. **Add a cell**
   Click "+ Code" or "+ Markdown" in the toolbar
4. **Run a cell**
   Click ▶ or press Shift + Enter
5. **Run all cells**
   Kernel → Restart & Run All (clears old outputs first)

Markdown in Jupyter Notebooks (Cheat Sheet here!)

Syntax	Renders as
`# Heading 1`	**Large section title**
`## Heading 2`	**Subsection title**
`**bold text**`	**bold text**
`*italic text*`	*italic text*
`- item\n- item`	• Bulleted list
`1. first\n2. sec`	1. Numbered list
`` `inline code` ``	`monospace snippet`

What is a CSV?

A .csv file stores tabular data as plain text. Each line is a row, and values in that row are separated by commas.

column

column headers

row of data

```
year,temp,result
2018,48.24,no shadow
2019,46.63,shadow
2020,52.50,shadow
2021,45.86,no shadow
2022,50.36,shadow
```

What is a CSV?

A .csv file stores tabular data as plain text. Each line is a row, and values in that row are separated by commas.

In your VSCode workspace:

1. In your **`data_analysis`** directory, create a file called **`walter_wally.csv`**
2. Copy the data to the right, and paste it into **`walter_wally.csv`**
3. Save the .csv file

```
year,temp,result
2018,48.24,no shadow
2019,46.63,shadow
2020,52.50,shadow
2021,45.86,no shadow
2022,50.36,shadow
```

Let’s write a function to help us read and wrangle the data!

`read_csv_rows`

`read_csv_rows` should read an entire CSV of data and return it as a list of rows, where each row represents a `dict[str, str]`.

- Function Name: `read_csv_rows`
- Parameter:
  - `str` path to CSV file
- Return Type: `list[dict[str, str]]`

Then, let’s call it to read our CSV data and convert them to a `list[dict[str, str]]`

read_csv_rows

Before calling read_csv_rows (data in our .csv):

```
year,temp,result
2018,48.24,no shadow
2019,46.63,shadow
2020,52.50,shadow
2021,45.86,no shadow
2022,50.36,shadow
```

After calling read_csv_rows (Python objects in our .ipynb):

```
[{'year': '2018', 'temp': '48.24', 'result': 'no shadow'}
{'year': '2019', 'temp': '46.63', 'result': 'shadow'},
{'year': '2020', 'temp': '52.50', 'result': 'shadow'},
{'year': '2021', 'temp': '45.86', 'result': 'no shadow'},
{'year': '2022', 'temp': '50.36', 'result': 'shadow'}]
```

`read_csv_rows` (solution)

```python
def read_csv_rows(filename: str) -> list[dict[str, str]]:
    """Read the rows of a CSV into a 'table'."""
    result: list[dict[str, str]] = []

    # Open a handle to the data file
    file_handle = open(filename, "r", encoding="utf8")

    # Prepare to read the data file as a CSV rather than just strings.
    csv_reader = DictReader(file_handle)

    # Read each row of the CSV line-by-line
    for row in csv_reader:
        result.append(row)

    # Close the file when done, to free its resources.
    file_handle.close()

    return result
```

The **`read_csv_rows`** function we just wrote together will be one of nine functions provided in EX09.

The other 8 will be provided to you, along with descriptions (“documentation”).

This mirrors real-world development: using functions written by others and interpreting their code and documentation to use them correctly.

Let’s read through the code and documentation of another function that will be provided to you in EX09! →

`column_values` function definition

```python
def column_values(table: list[dict[str, str]], column: str) -> list[str]:
    """Produce a list[str] of all values in a single column."""
    result: list[str] = []

    for row in table:
        item: str = row[column]
        result.append(item)

    return result
```

With a neighbor, discuss:

1. Consider the **`read_csv_rows`**' and **`column_values`**' function signatures. What do they have in common?
    a. Could the return value of one function call be an argument in the other's function call?
2. Inside the for loop, what type of data would **`row`** store?
3. Does the literal value of **`column`** ever change?

`column_values` function definition and call

```
def column_values(table: list[dict[str, str]], column: str) -> list[str]:
    """Produce a list[str] of all values in a single column."""
    result: list[str] = []

    for row in table:
        item: str = row[column]
        result.append(item)

    return result

pairs: list[dict[str, str]] = [{"owner": "Alyssa", "pet": "Bear"}, {"owner":
"Izzi", "pet": "Pip"}, {"owner": "Kaki", "pet": "Lily"}]

print(column_values(pairs, "pet"))
```

With a neighbor, discuss:

1. What is the output of this code listing?

`column_values` function definition and call

```
def column_values(table: list[dict[str, str]], column: str) -> list[str]:
    """Produce a list[str] of all values in a single column."""
    result: list[str] = []

    for row in table:
        item: str = row[column]
        result.append(item)

    return result

data_rows: list[dict[str, str]] = read_csv_rows(<your/path/to/walter_wally.csv>)

years: list[str] = column_values(data_rows, "year")
```

With a neighbor, discuss:

1. What is the length of **`years`**?
2. What list literal does **`years`** store?
3. Where would the variable name **`years`** and its contents be written in a memory diagram, respectively?

`column_values` function definition & documentation

```python
def column_values(table: list[dict[str, str]], column: str) -> list[str]:
    """Produce a list[str] of all values in a single column."""
    result: list[str] = []

    for row in table:
        item: str = row[column]
        result.append(item)

    return result
```

- Purpose: Produce a `list[str]` of all values in a single `column` whose name is the second parameter.
- Function Name: `column_values`
- Parameters:
    1. `list[dict[str, str]]` - a list of rows representing a _table_
    2. `str` - the name of the column (key) whose values are being selected
- Return Type: `list[str]`

`columnar` function definition

```python
def columnar(row_table: list[dict[str, str]]) -> dict[str, list[str]]:
    """Transform a row-oriented table to a column-oriented table."""
    result: dict[str, list[str]] = {}

    first_row: dict[str, str] = row_table[0]
    for column in first_row:
        result[column] = column_values(row_table, column)

    return result
```

- Purpose: *Transform* a table represented as a list of rows (e.g. `list[dict[str, str]]`) into one represented as a dictionary of columns (e.g. `dict[str, list[str]]`).
- Function Name: `columnar`
- Parameter: `list[dict[str, str]]` - a "table" organized as a list of rows
- Return Type: `dict[str, list[str]]` - a "table" organized as a dictionary of columns

`columnar` function purpose

Before calling `columnar`:

```
[{'year': '2018', 'temp': '48.24', 'result': 'no shadow'}
{'year': '2019', 'temp': '46.63', 'result': 'shadow'},
{'year': '2020', 'temp': '52.50', 'result': 'shadow'},
{'year': '2021', 'temp': '45.86', 'result': 'no shadow'},
{'year': '2022', 'temp': '50.36', 'result': 'shadow'}]
```

After calling `columnar`:

```
{'year': ['2018', '2019', '2020', '2021', '2022'],
'temp': ['48.24', '46.63', '52.50', '45.86', '50.36'],
'result': ['no shadow', 'shadow', 'shadow', 'no shadow', 'shadow']}
```

With a partner, discuss:

1. What are some of the benefits/drawbacks of column-wise data (in general, and in this example in Python)?